

Supervised Learning

Module 3.1 — Machine Learning Fundamentals

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Agenda

Here is what we will cover in this module on Supervised Learning:

01

Introduction to Supervised Learning

03

Applications Across Industries

02

Categories: Regression & Classification

04

Key Algorithms (Linear Regression, Logistic Regression, Naïve Bayes, KNN, Decision Trees, Random Forests, SVM)

What is Supervised Learning?

Supervised learning is a type of machine learning where the algorithm **learns from labeled data** — input-output pairs. The algorithm is trained on a dataset that includes input features and correct output labels.

The goal is for the algorithm to learn the **mapping between inputs and outputs**, so it can make accurate predictions on new, unseen data.

- ❏ Think of it like a student learning from a textbook with answers — the model studies examples to predict future outcomes.

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Supervised Learning Categories

There are **two main categories** of supervised learning, based on the type of output (target variable):

Regression

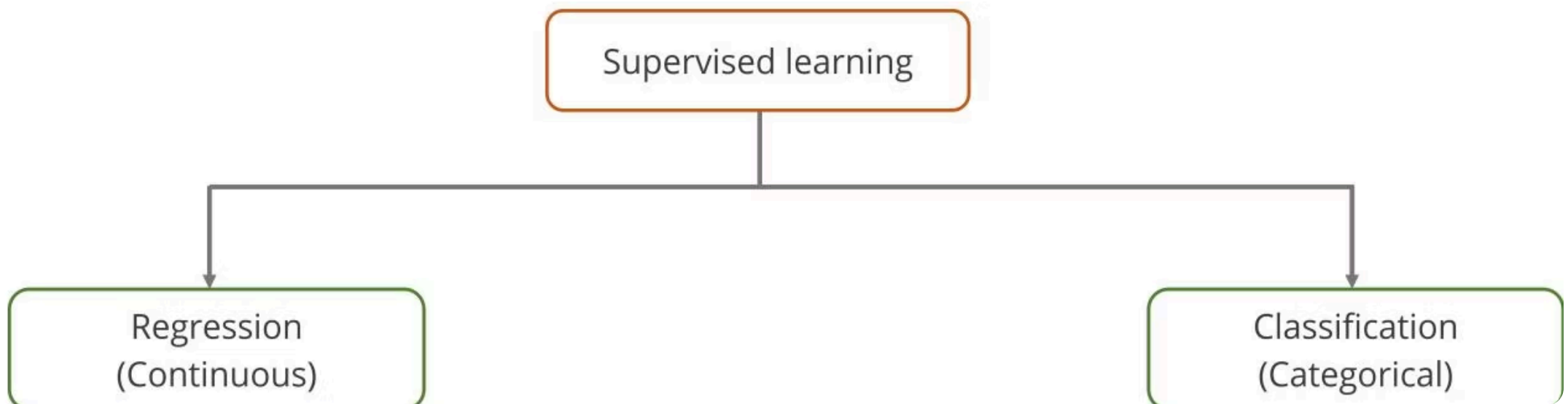
Output is continuous. Predicts a specific numeric value.

Example: Estimating the price of a house based on size, location, and age.

Classification

Output is categorical. Assigns data to a class or group.

Example: Predicting whether an email is **spam** or **not spam**.

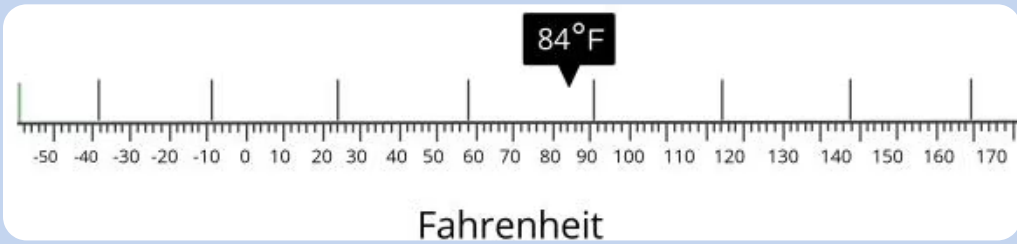


Regression vs. Classification — A Weather Example

Regression

Question: What will be the temperature tomorrow?

Output: A specific number — e.g., **84°F**

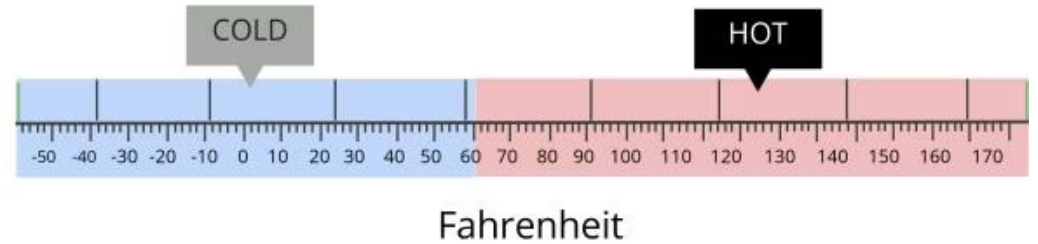


Predicts a **continuous value** on a scale.

Classification

Question: Will it be hot or cold tomorrow?

Output: A category — **HOT** or **COLD**



Predicts a **categorical label** from defined classes.

Applications of Supervised Learning

Supervised learning **optimizes and automates processes** across many industries, enhancing decision-making and reducing manual effort.



HR Operations

Shortlists resumes based on accepted/rejected candidate profiles — like Netflix recommending movies.



Finance

Automates loan approval and risk assessment using historical loan data.



Emails

Spam filters use supervised learning to classify emails as spam or not spam.



Manufacturing

Inspects product quality and classifies items into grades, detecting defects automatically.

More Applications

Maritime Industry

Regression models use historical weather and event data to predict tidal currents, forecast demand and supply, and reduce inventory losses — like a weather forecast predicting rain.

Fraud Protection

Models detect fraudulent financial transactions, cyber attacks, and fake social media profiles by learning from past fraud instances — like a detective learning from previous cases.

📄 Supervised learning enables **more accurate predictions** and **smarter automation** across every sector.

Supervised Learning Algorithms

An **algorithm** is the specific method used to train a model on labeled data. Here are the most commonly used supervised learning algorithms:



Linear Regression

Predicts housing prices using features like bedrooms, size, and location.



Logistic Regression

Predicts if a customer will purchase a product based on browsing and demographics.



Naïve Bayes

Classifies emails as spam or not spam based on words and phrases used.



K-Nearest Neighbors

Predicts heart disease risk by comparing a patient's indicators with historical data.

More Algorithms

1

Decision Trees

Use case: Urban planning. Analyzes population density, transport usage, and community feedback to optimize public transportation — a structured, rule-based approach.

2

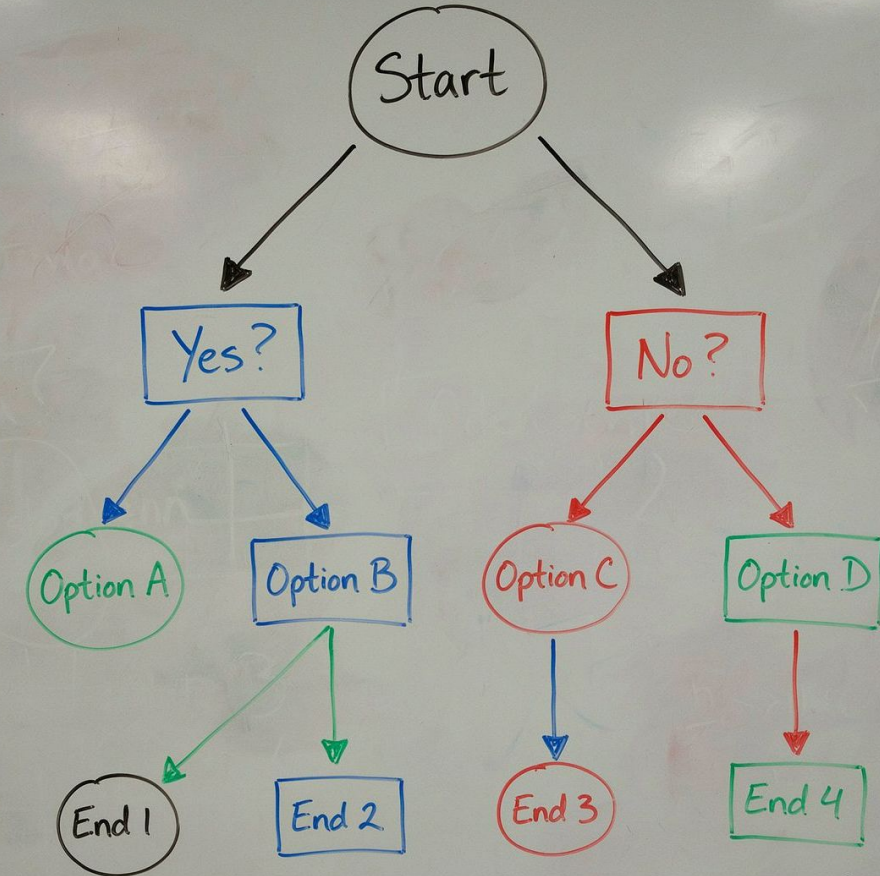
Random Forests

Use case: Credit scoring. A bank uses multiple decision trees together to evaluate creditworthiness — income, credit history, employment — for more accurate loan decisions.

3

Support Vector Machines (SVM)

Use case: Social media monitoring. Classifies customer comments as **positive** or **negative** to gauge public perception and support brand management.





Key Takeaways

- Supervised learning trains on labeled input-output data to predict outcomes on new data.
- Two categories: **Regression** (continuous output) and **Classification** (categorical output).
- Applied across HR, Finance, Email, Manufacturing, Maritime, and Fraud Detection.
- Key algorithms: Linear Regression, Logistic Regression, Naïve Bayes, KNN, Decision Trees, Random Forests, SVM.

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